AUTOMATIC SOAP DISPENSER

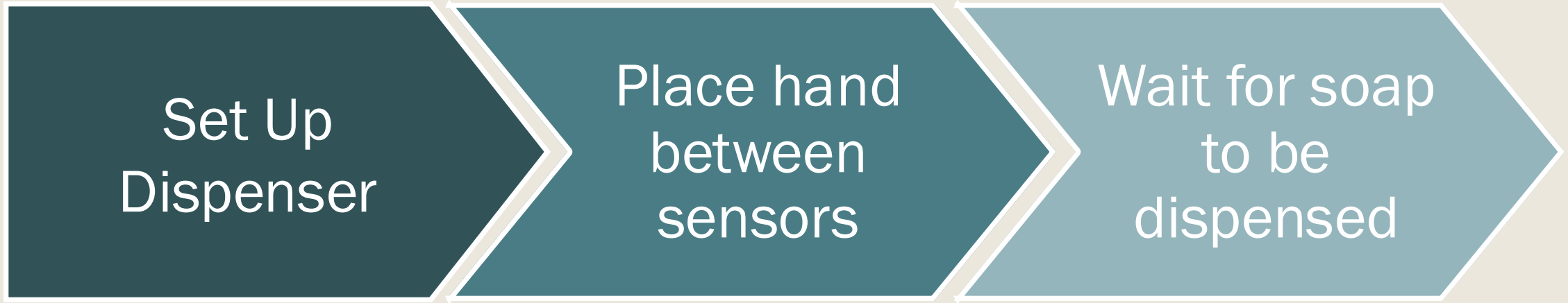
Joseph Arens, Faith Godden, Jacob Yehuda, Brett Womack

Executive Overview

- An Automatic Soap Dispenser
- For use in EPIC
- People whose hands may be dirty
- Main objectives: Release Soap, Touch Free



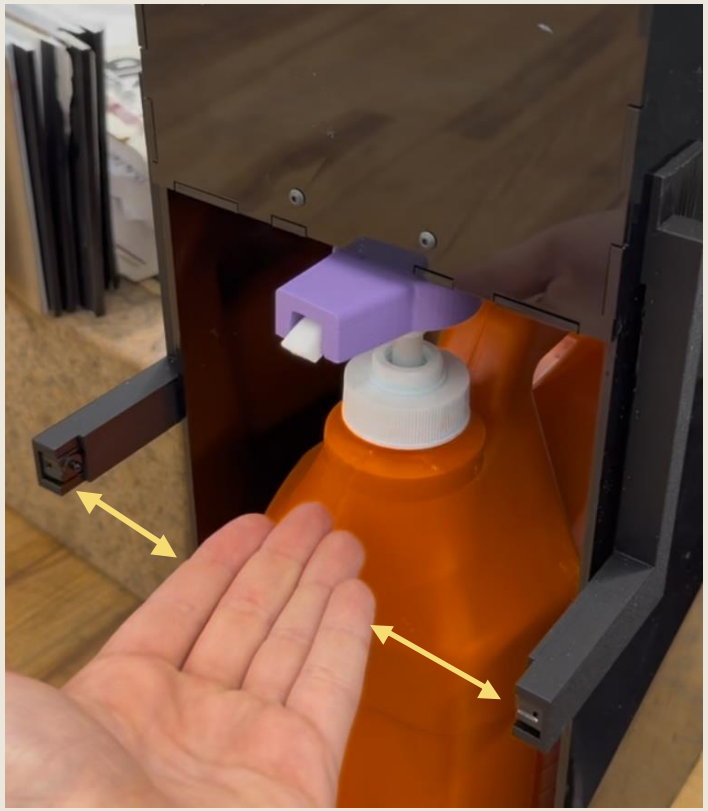
Description of Usability



Key Point – No Contact

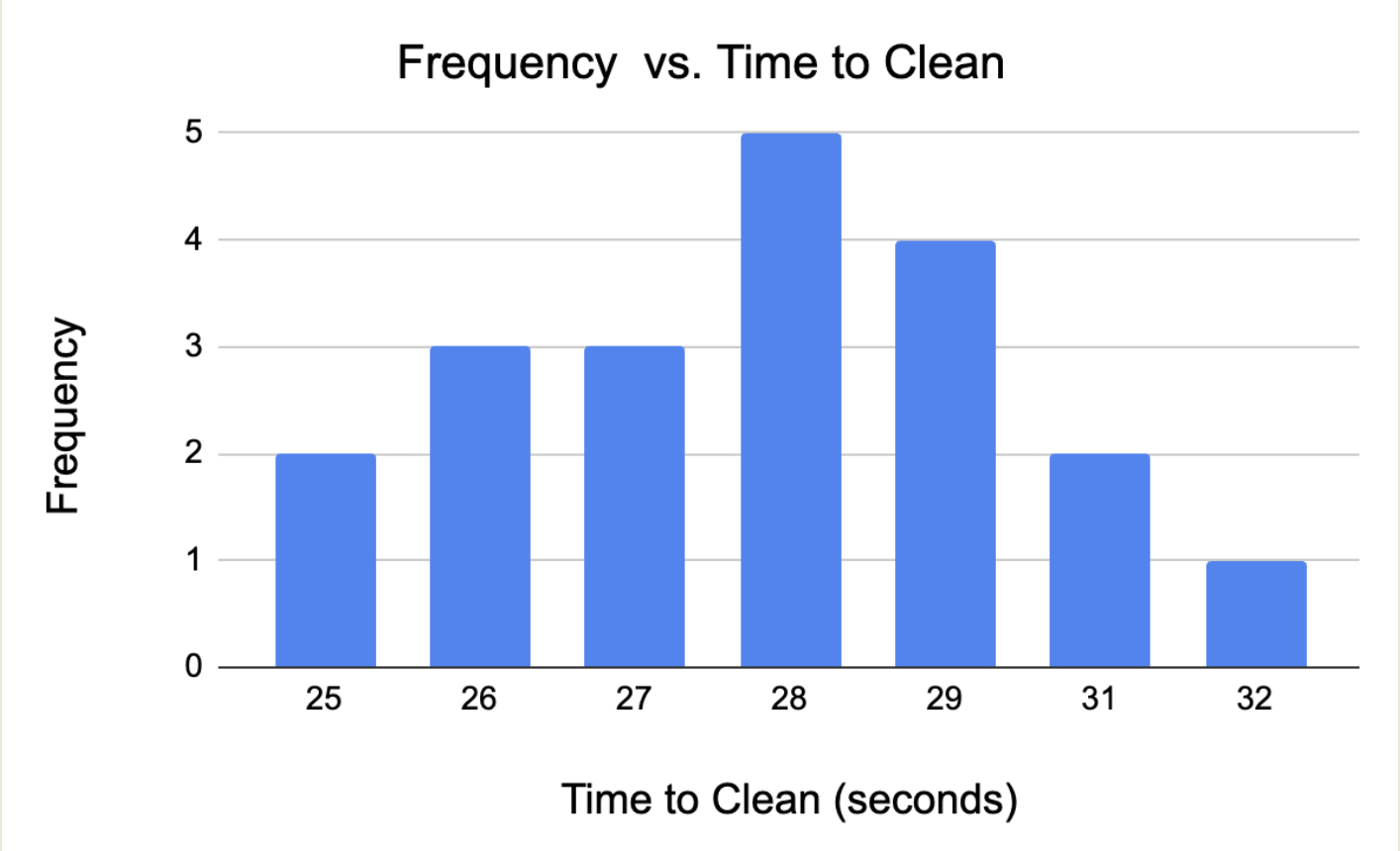
- More than one method of use
- **Metric:** No contact (x>1cm)
- **Performance:**
  - Perpendicular: 2cm clearance
  - Parallel: 5cm clearance





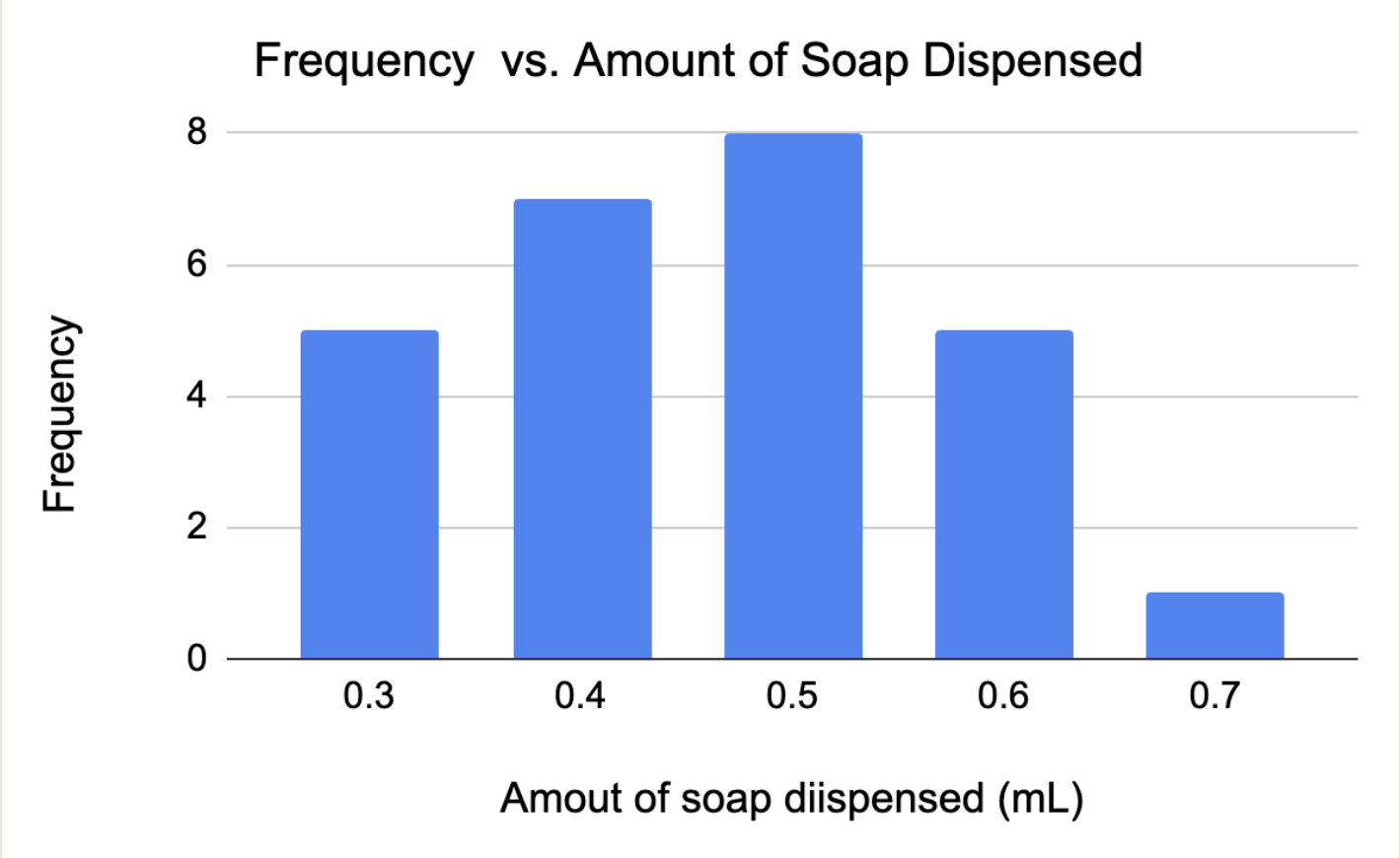
Key Point – Easy to Clean

- **Metric:** Clean in less than 60s
- **Results:** Normal distribution centered around 28s

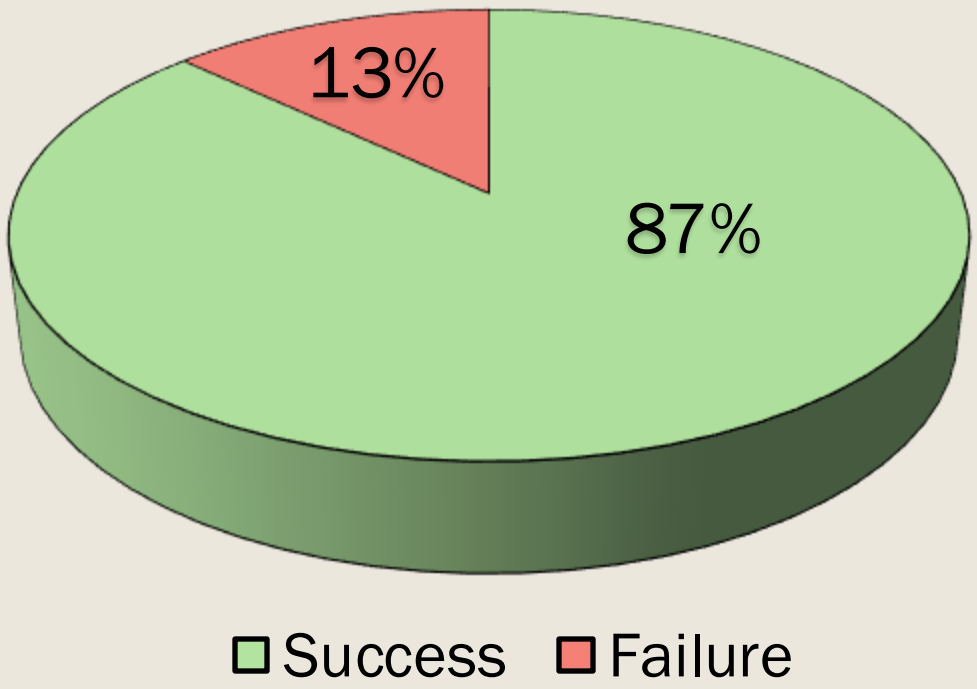


Key Point – Amount Pumped

- **Metric:** x>0 soap
- **Results:** Average of 0.5mL
- About 7570 pumps per GOJO bottle

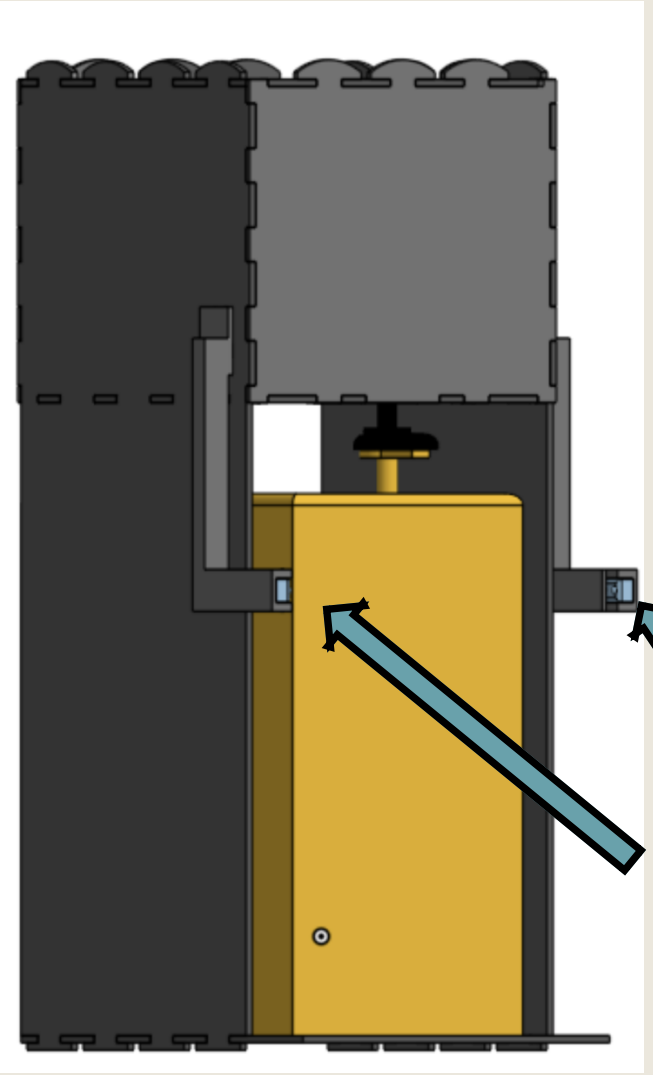


Key Point – Reliability



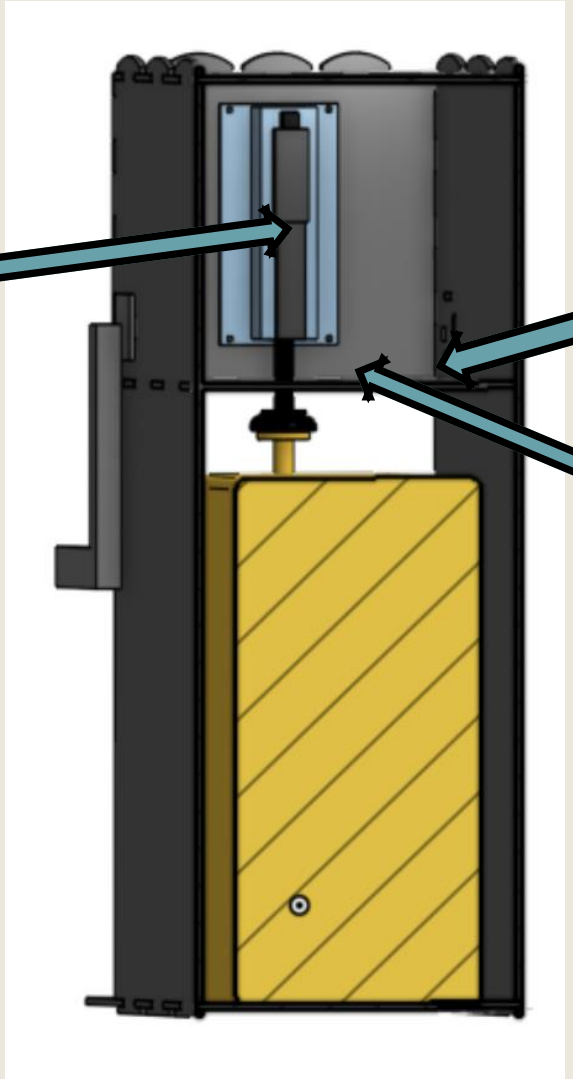
- **Metric:** 96% reliable
- 30 tests performed
- Pump became stiffer
- Improvements: stronger linear actuator

Pictures of Key Parts

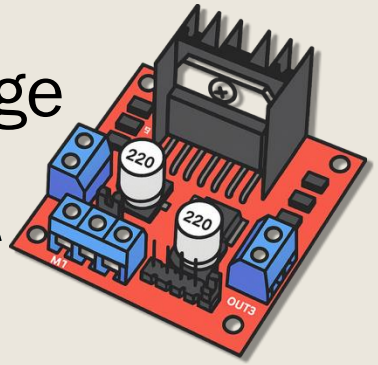


IR Break Beam Sensor

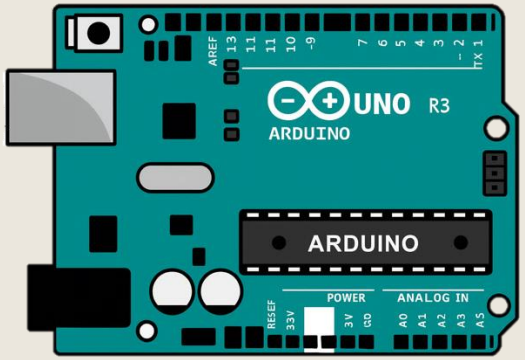
Linear Actuator



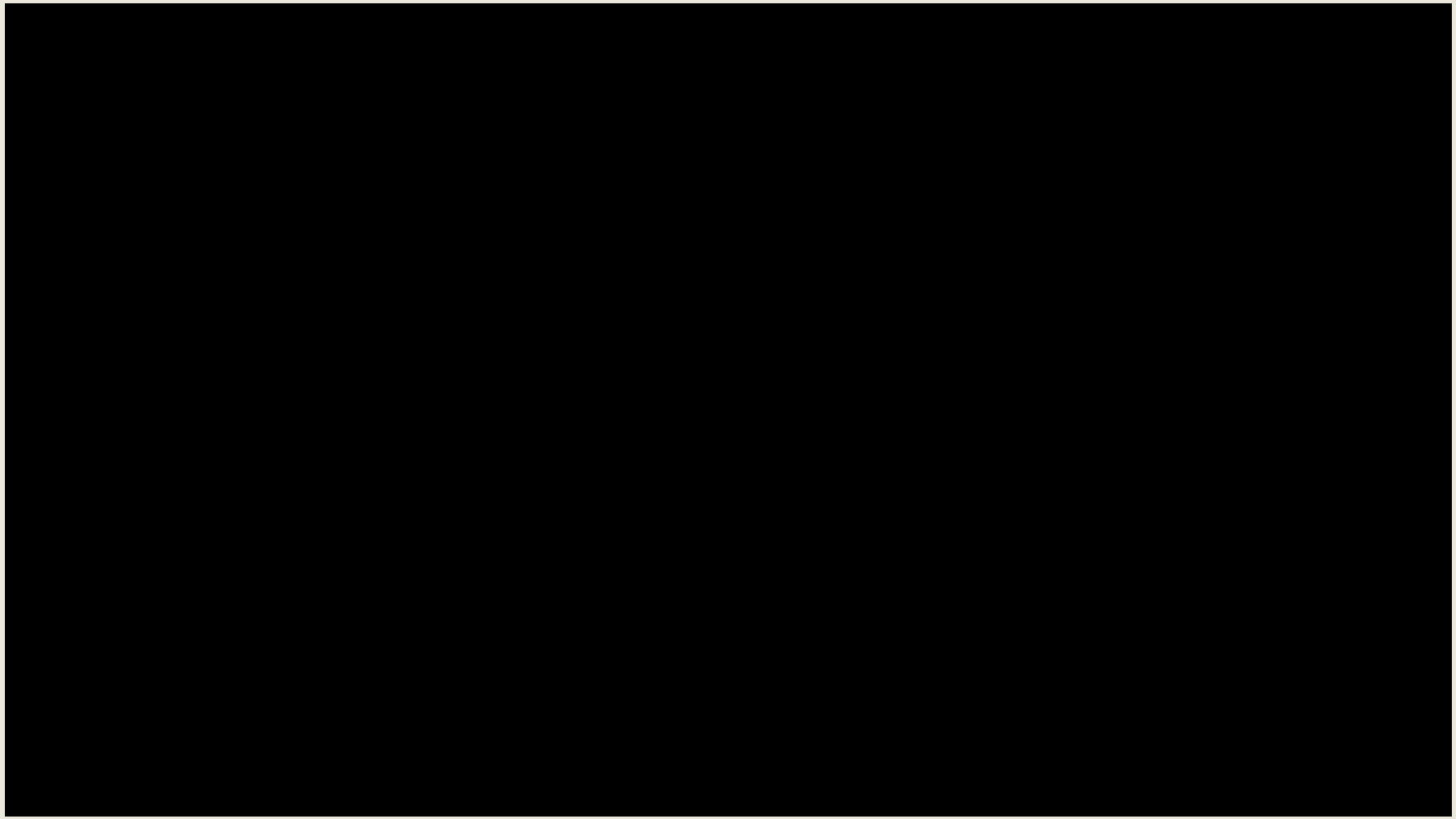
H-Bridge



Arduino Uno



Video



THANK YOU

From Team Parlay